

10 Backups

We will now discuss the process of creating snapshot style backups. Snapshot backups are a feature of some high-end industrial file servers; they create the illusion of multiple, full backups per day without the space or processing overhead. It is often possible to store several weeks worth of snapshots with slightly more than twice the original storage. We will stick to the basics and only use file utilities and another tool called `rsync`. The advantage is obvious enough, you can revert to a snapshot in case of data loss or a miscalculated configuration change.

10.1 Using `rsync` to make a backup

`Rsync` is your swiss army knife for backups, not only is it full of features, it is also very polished and powerful. Let us learn it in detail, you will find its applications in areas even other than backup, including monitoring resources.

Covering the obvious parts first, assume you have a directory called `source`, and you want to back it up into the directory `destination`.

In the most simple manner, you will just need to issue the following

```
$ rsync -a source/ destination/
```

you can add `-v` or `-vv` to adjust the verbosity of the output according to you, at this point you might be wondering how is this any different from the following:

```
$ cp -a source/. Destination/
```

Now as a matter of fact there won't be any difference in the first run, but following that only changes get updated, so you can sync say 100 GB of records in which only a few MB change everyday in 2 minutes or less resulting in high efficiency.

We have covered `ssh` in great detail above, what is the point if we don't apply it here, so:

```
rsync -a -e ssh source/ username@remotemachine.com:/  
path/to/destination/
```

10.2 Confusion around trailing slashes

You may be accustomed to commands that don't care about trailing slashes. For example, if `a` and `b` are two directories, then `cp -a a b` is equivalent to `cp -a a/ b/`. However, `rsync` does care about the trailing slash, but only on the source argument. For example, let `a` and `b` be two directories, with the file `foo` initially inside directory `a`. Then type this command:

```
rsync -a a b
```